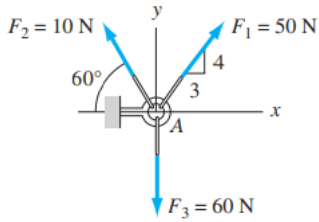
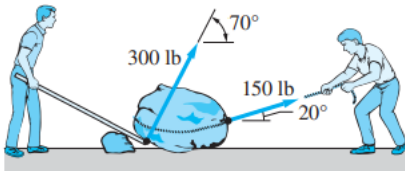


Resultant of Concurrent forces

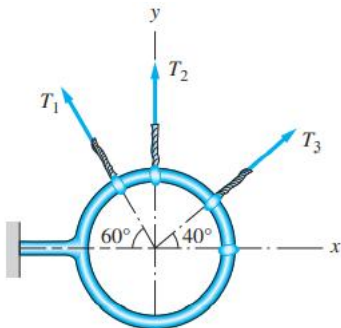
1. Determine the resultant of the three concurrent forces shown in Fig.



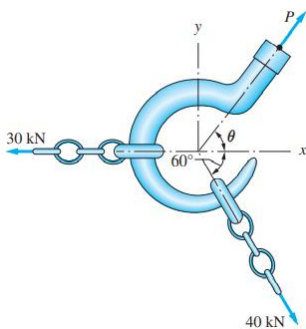
2. Two men are trying to roll the boulder by applying the forces shown in figure. Find the magnitude and direction of the force that is equivalent to the two applied forces.



3. The magnitudes of the three forces applied to the eye bolt are $T_1 = 110$ N, $T_2 = 40$ N, and $T_3 = 150$ N. Replace these forces with a single equivalent force R . Show the result on a sketch of the eye bolt.



4. Determine P and θ so that the three forces shown are equivalent to the single force $R = 85i + 20j$ kN

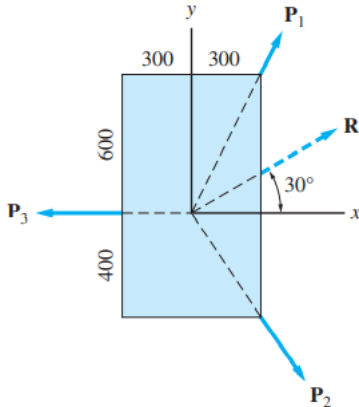


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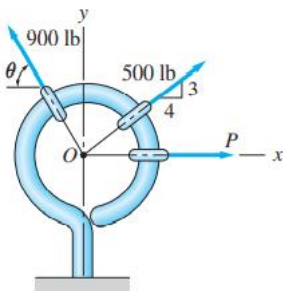
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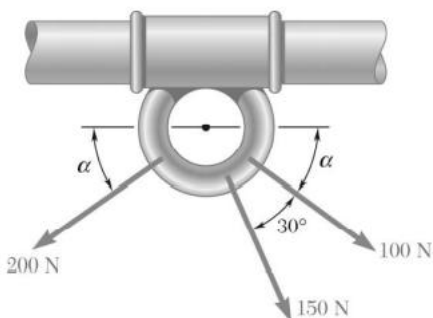
5. The force R is the resultant of the forces P_1 , P_2 , and P_3 acting on the rectangular plate. Find P_1 and P_2 if $R = 40$ kN and $P_3 = 20$ kN.



6. The resultant of the three concurrent forces acting on the eyebolt is the force $R = 800\mathbf{j}$ lb. Determine the magnitude of the force P and the angle θ that specifies the direction of the 900-lb force.



7. For the collar shown in figure, determine (a) the required value of α if the resultant of the three forces shown is to be vertical, (b) the corresponding magnitude of the resultant.

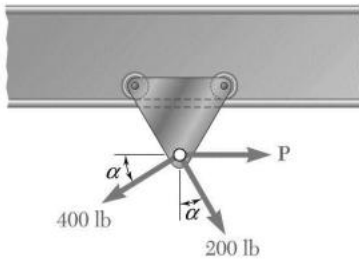


8. A hoist trolley is subjected to the three forces shown. Knowing that $\alpha = 40^\circ$, determine (a) the required magnitude of the force P if the resultant of the three forces is to be vertical, (b) the corresponding magnitude of the resultant.

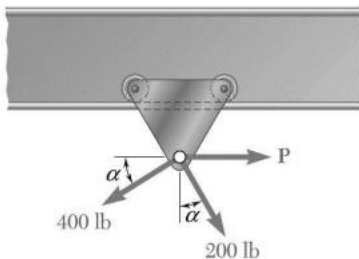
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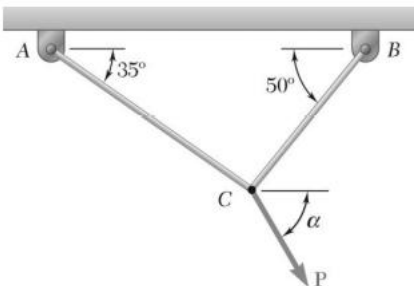
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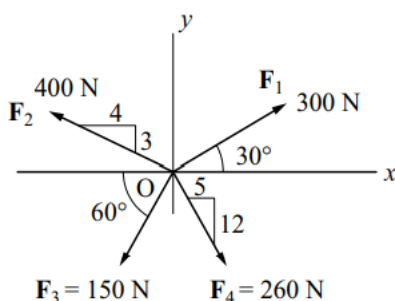
9. A hoist trolley is subjected to the three forces shown. Knowing that $P = 250$ lb, determine (a) the required value of α if the resultant of the three forces is to be vertical, (b) the corresponding magnitude of the resultant.



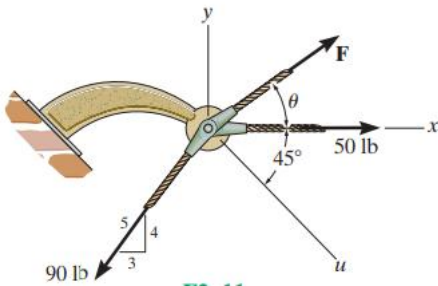
10. Two cables tied together at C are loaded as shown. Knowing that the maximum allowable tension in each cable is 800 N, determine (a) the magnitude of the largest force P that can be applied at C, (b) the corresponding value of α .



11. Determine the resultant of four forces acting on a body as shown below

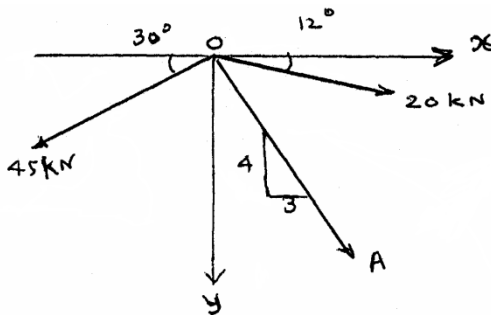


12. If the magnitude of the resultant force acting on the bracket is to be 80 lb directed along the u axis, determine the magnitude of F and its direction.



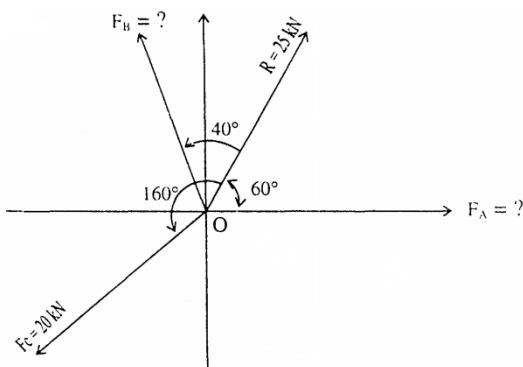
13. Find the force in the cable OA , if the resultant of the forces at O is acting vertically. What is the magnitude of the resultant?

(Ans: $T_{OA} = 32.35$ kN, $R = 52.54$ kN)



14. A force $R = 25$ kN acting at O has three components, F_A , F_B and F_C as shown in fig. If $F_C = 20$ kN, find F_A & F_B .

(Ans: $F_A = 33.9$ kN, $F_B = 35.03$ kN)



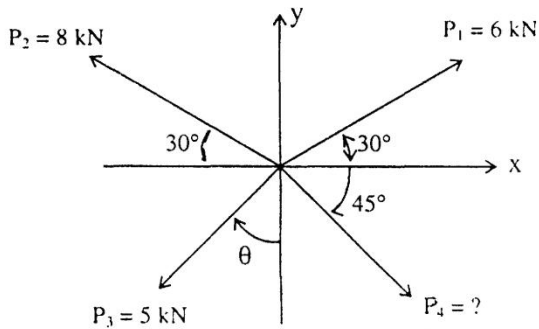
15. Find out the magnitude of the force P_4 and the direction of force P_3 if the resultant of four coplanar concurrent forces P_1 , P_2 , P_3 & P_4 as shown in fig. is zero.

(Ans: $P_4 = 9.51$ kN, $\theta = 87.13^\circ$)

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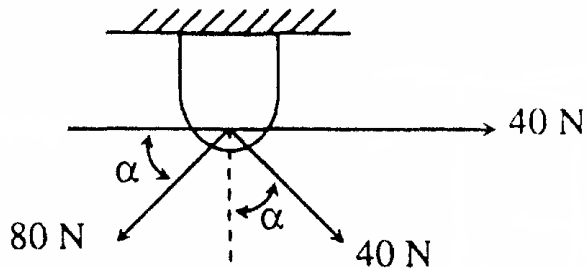
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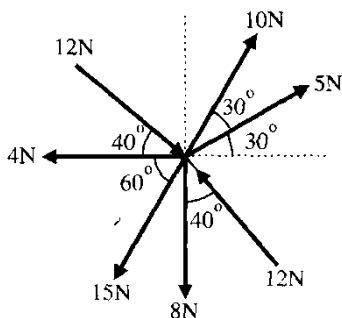
16. Three coplanar forces act at a point on a bracket as shown. Determine the value of the angle α such that the resultant of the three forces will be vertical. Also find the magnitude of the resultant.

(Ans: $\alpha = 36.87^\circ$, $R = 80 \text{ N}$)



17. Determine resultant of following force system as shown in Fig.

(Ans: $R = 8.38 \text{ N}$ IIIrd quadrant, $\theta = 85.28^\circ$)



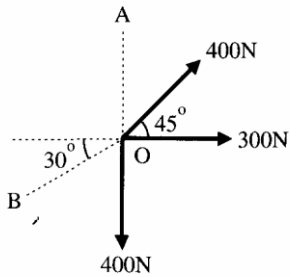
18. Three concurrent, coplanar forces act on a body at point o. Determine two additional forces, along OA and OB such that resultant of the five forces is zero.

(Ans: $F_{OA} = 453.66 \text{ N}$, $F_{OB} = 673 \text{ N}$)

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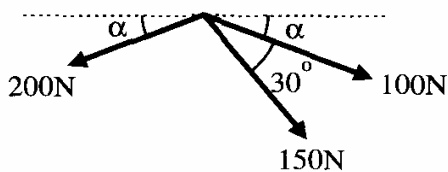
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19. For given system determine (a) the required value of α if resultant of 3 forces is to be vertical
 (b) The corresponding magnitude of resultant.

(Ans: $\alpha = 21.8^\circ$, $R = 229.29 \text{ N}$)



20. Determine the magnitude and direction of Force F_1 and F_2 ? When the Resultant of given force system is found to be 800 N along +ve x axis.

(Ans: $F_1 = 527.23$, $F_2 = 1161.2$)

